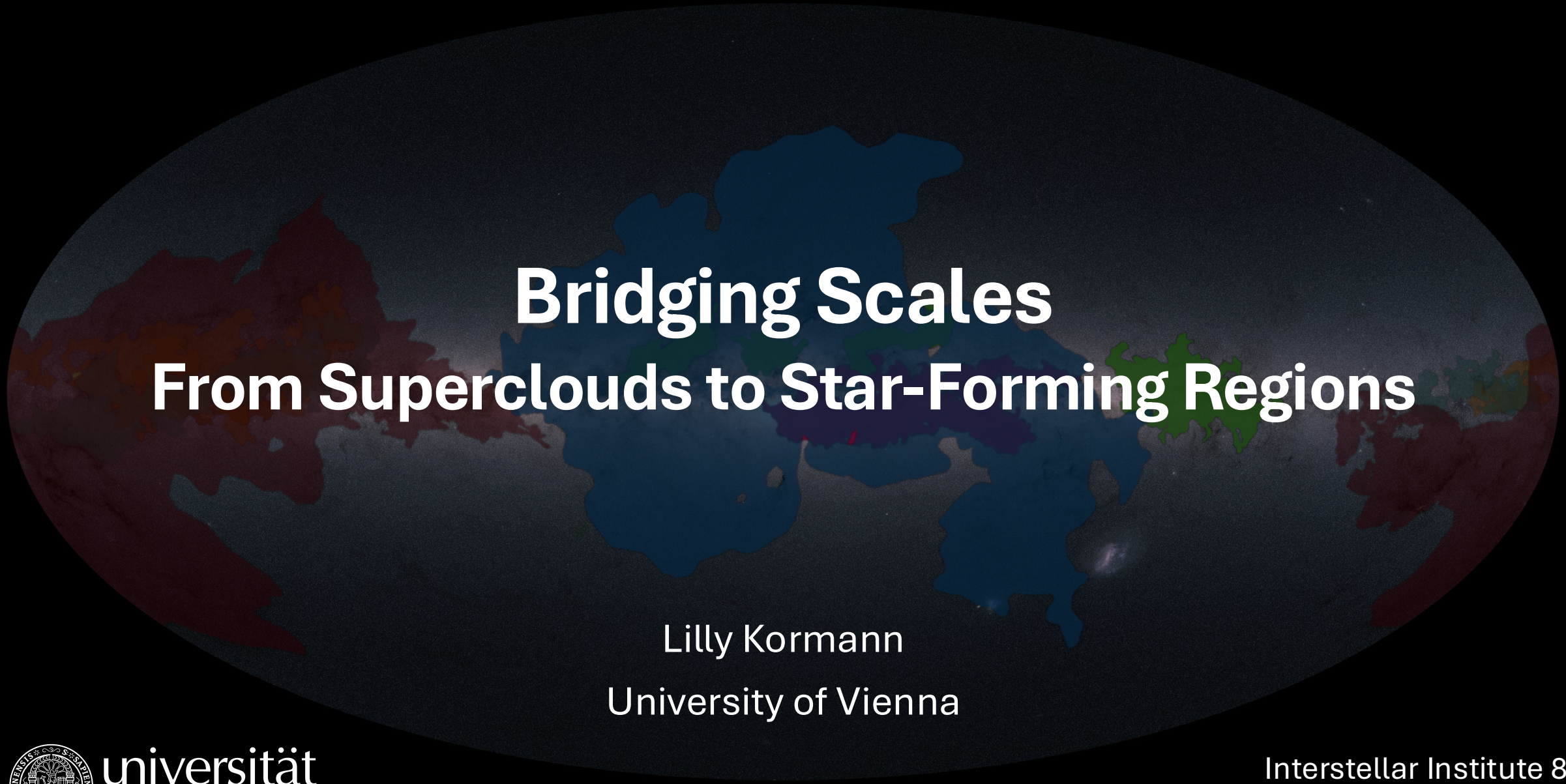


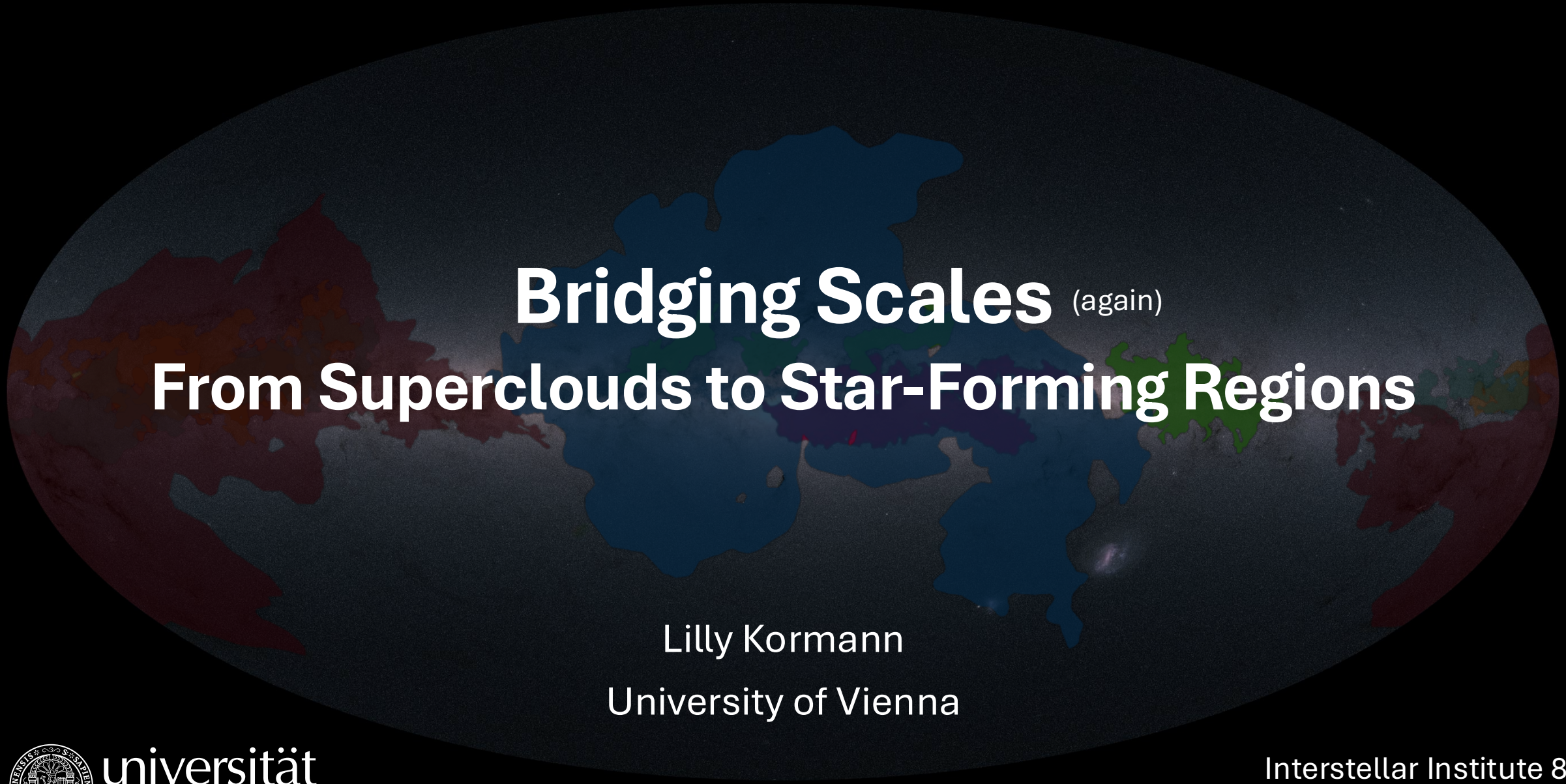
Bridging Scales
From Superclouds to Star-Forming Regions

Lilly Kormann

University of Vienna

universität wien

Interstellar Institute 8
28.09.2026



Bridging Scales (again)
From Superclouds to Star-Forming Regions

Lilly Kormann

University of Vienna

universität wien

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28.09.2026

Early History of Superclouds

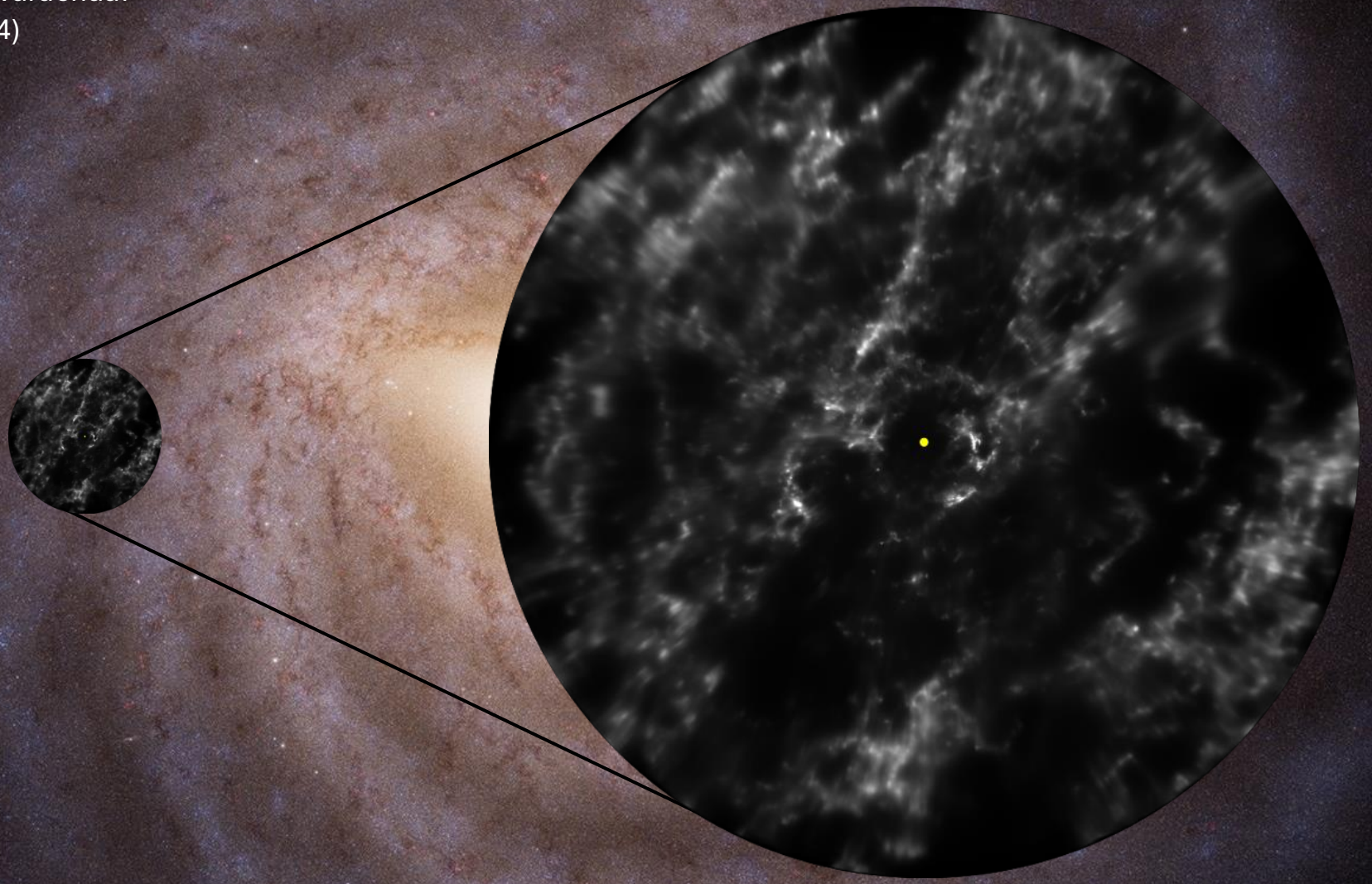
- some of the largest recognized structures within the interstellar medium

- first mentioned by Elmegreen & Elmegreen (1983)
  - neutral atomic hydrogen superclouds in spiral arms in external galaxies
  - masses between 10^6 and 10^7 $M_\odot$ and kiloparsec lengths

- further work done by Elmegreen & Elmegreen (1987)
  - identified neutral atomic hydrogen superclouds in the inner Galaxy
  - noted spatial correlation with giant molecular cloud complexes

Artistic Impression of the Milky Way

ESA/Gaia/DPAC, Stefan Payne-Wardenaar
Dust map: Edenhofer et al. (2024)



Animation by Efrem Maconi

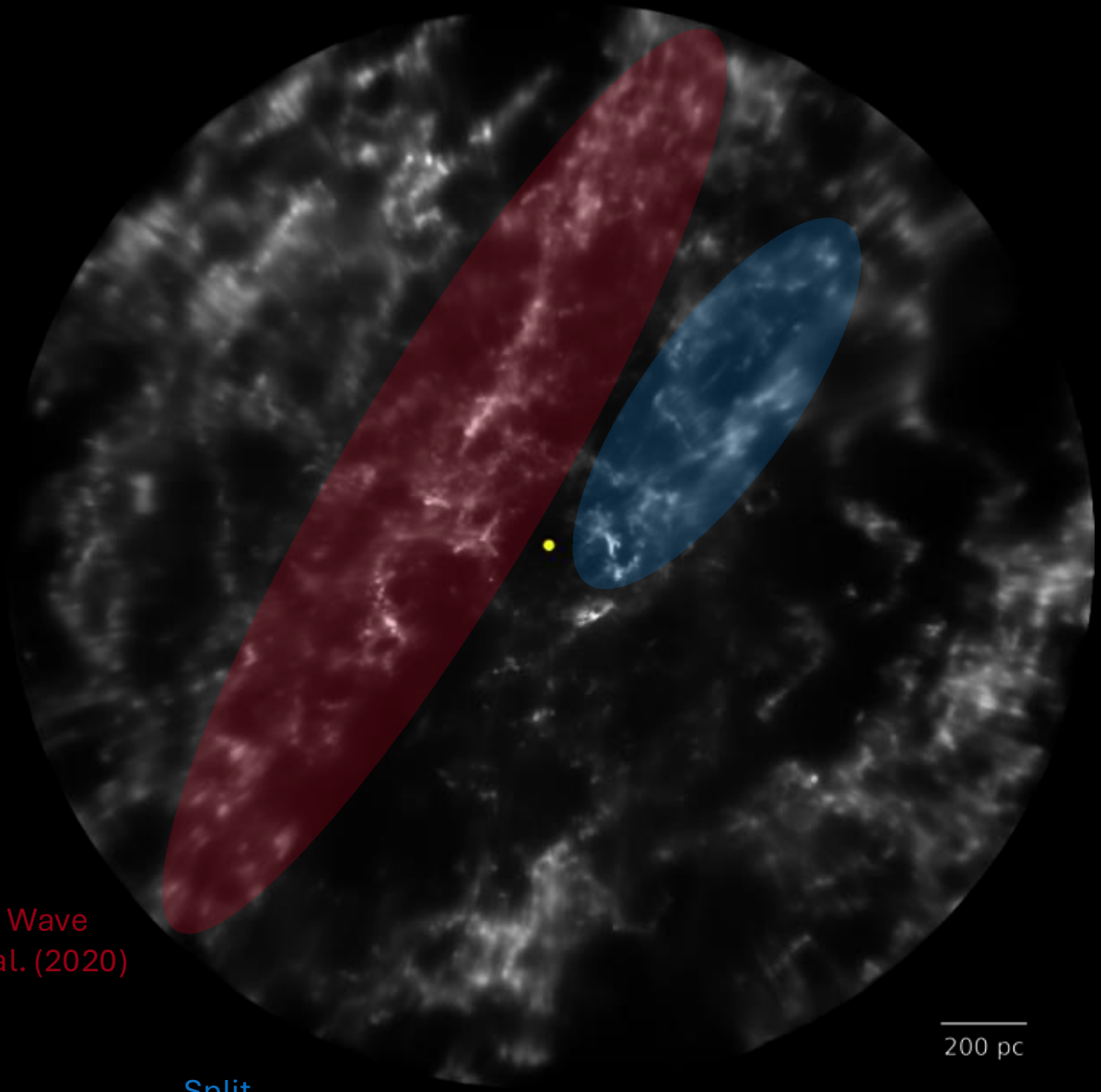
Finding Superclouds

previously known:

Radcliffe Wave

Split

HOP structure finder from Colman et al. (2024)

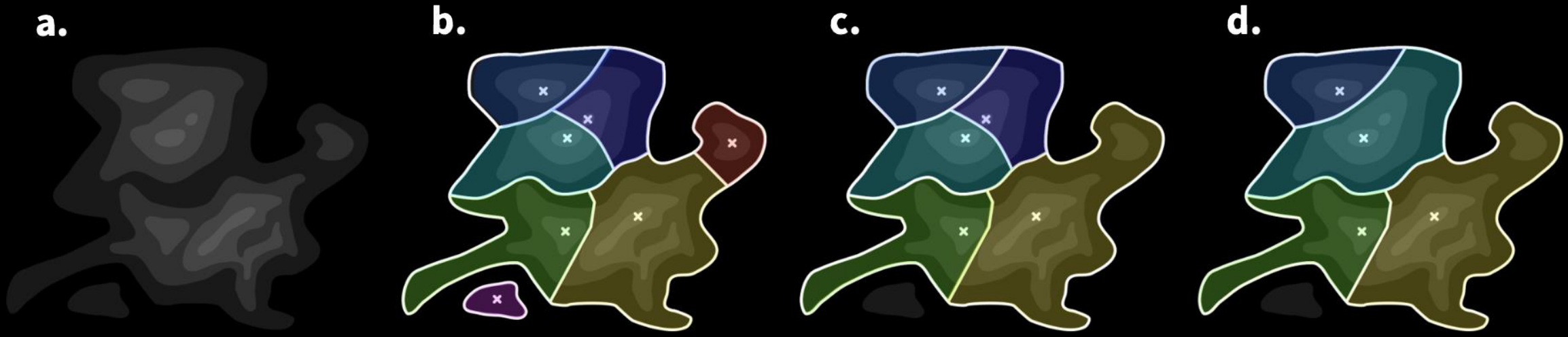


Radcliffe Wave
Alves et al. (2020)

Split
Lallement et al. (2019)

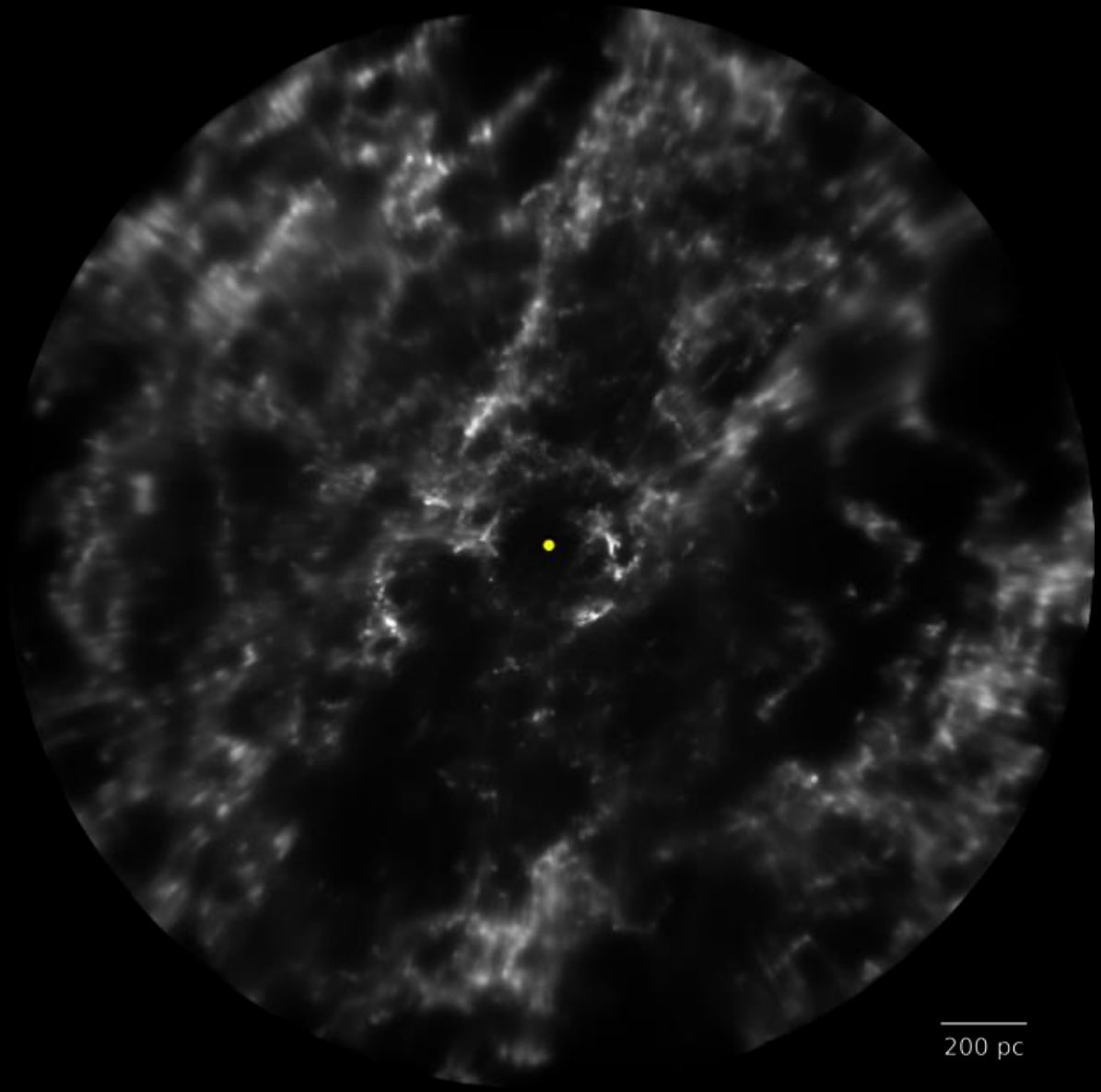
Structure Finding Algorithm HOP

- originally to identify groups in N-body simulations (Eisenstein & Hut 1998)
- wrapper by Colman et al. (2024) for MC identification in simulations
- analyzes voxels around peak patches and border densities for segmentation



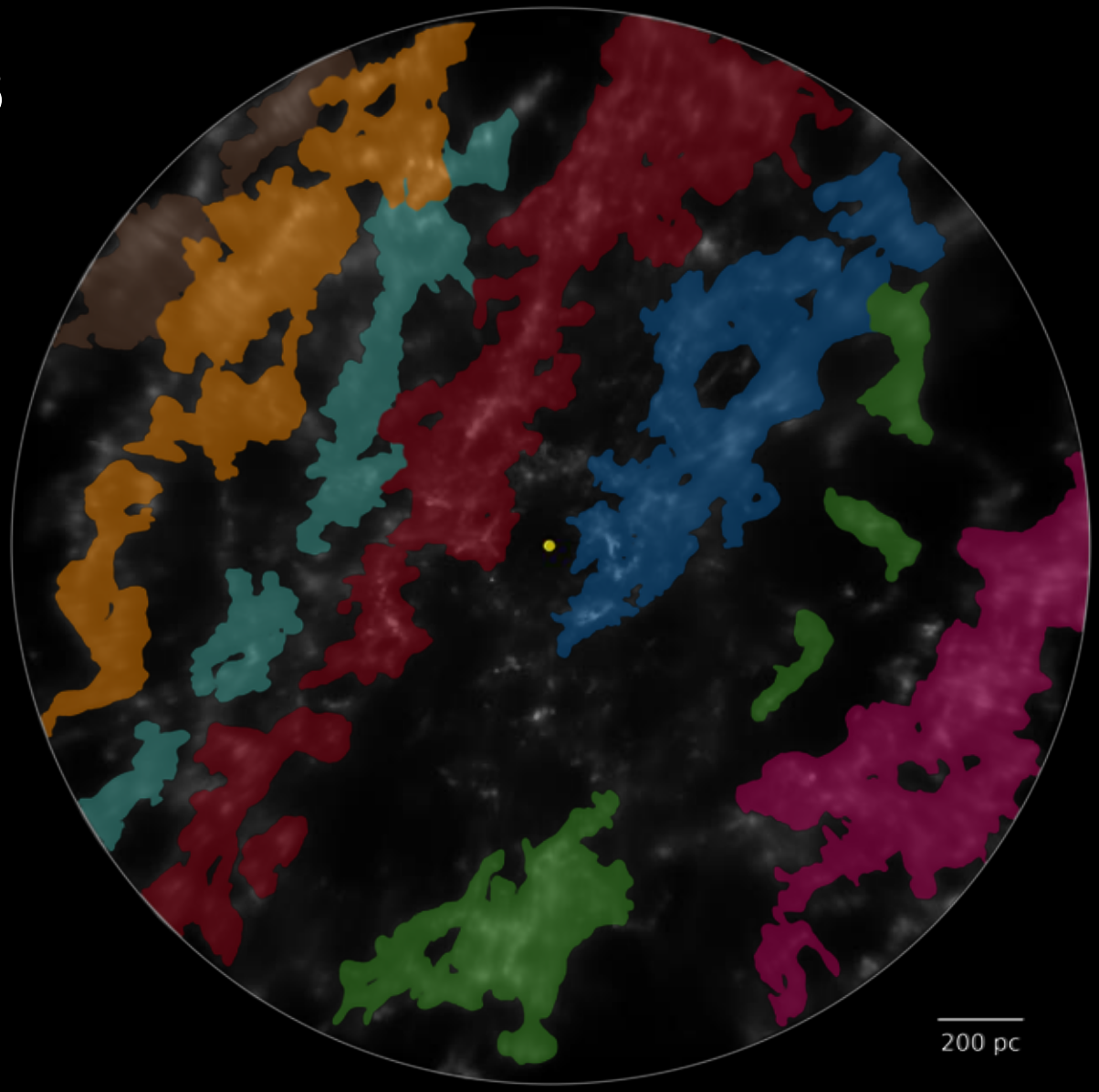
Finding Superclouds

> HOP structure finder from Colman et al. (2024)



Finding Superclouds

> HOP structure finder from Colman et al. (2024)



Superclouds

- 7 superclouds of the local Milky Way

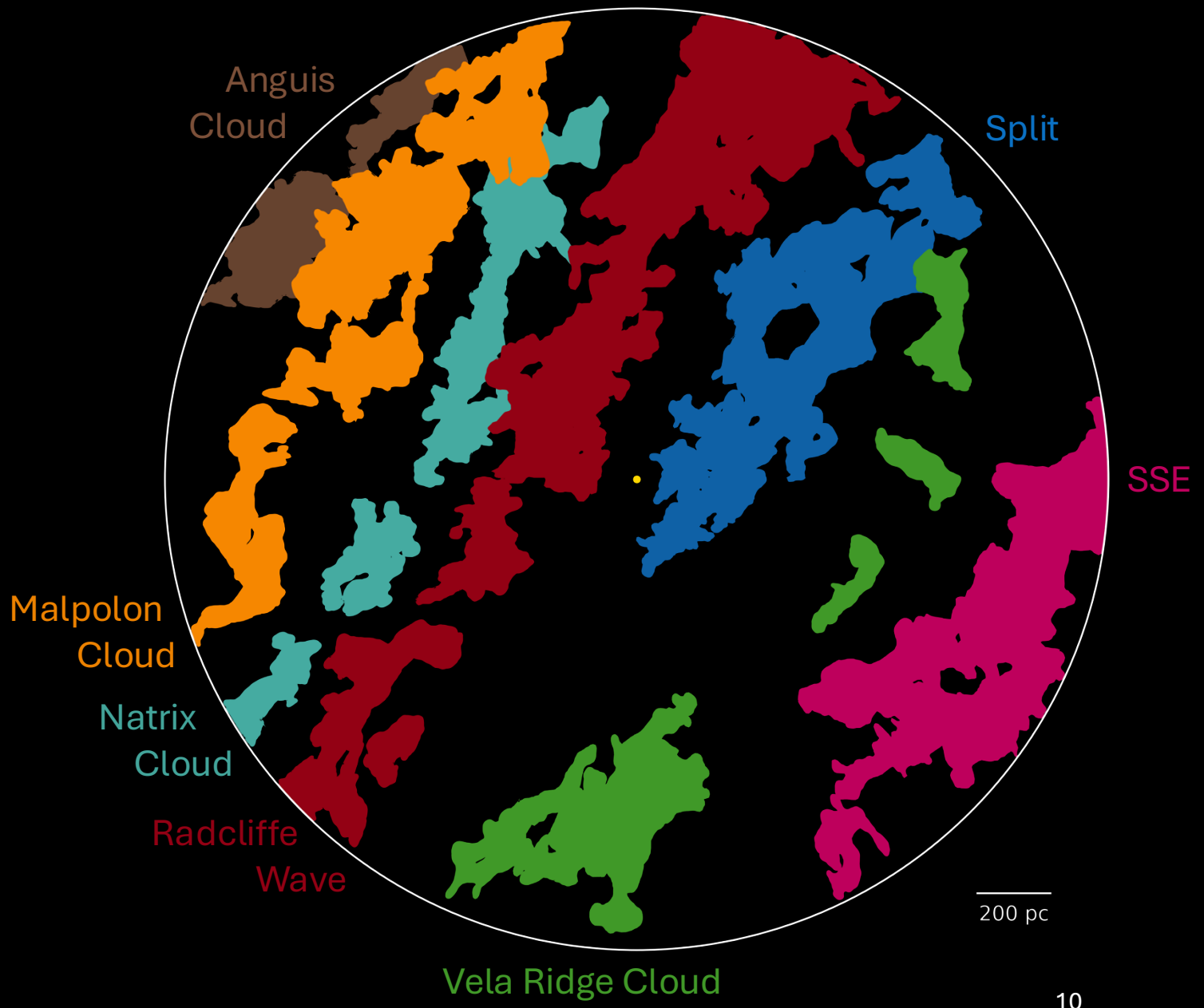
- lengths of 1 – 2.5 kpc
- masses of ~ 10^6 $M_\odot$
- ~ 30° pitch angle



Superclouds

- 7 superclouds of the local Milky Way

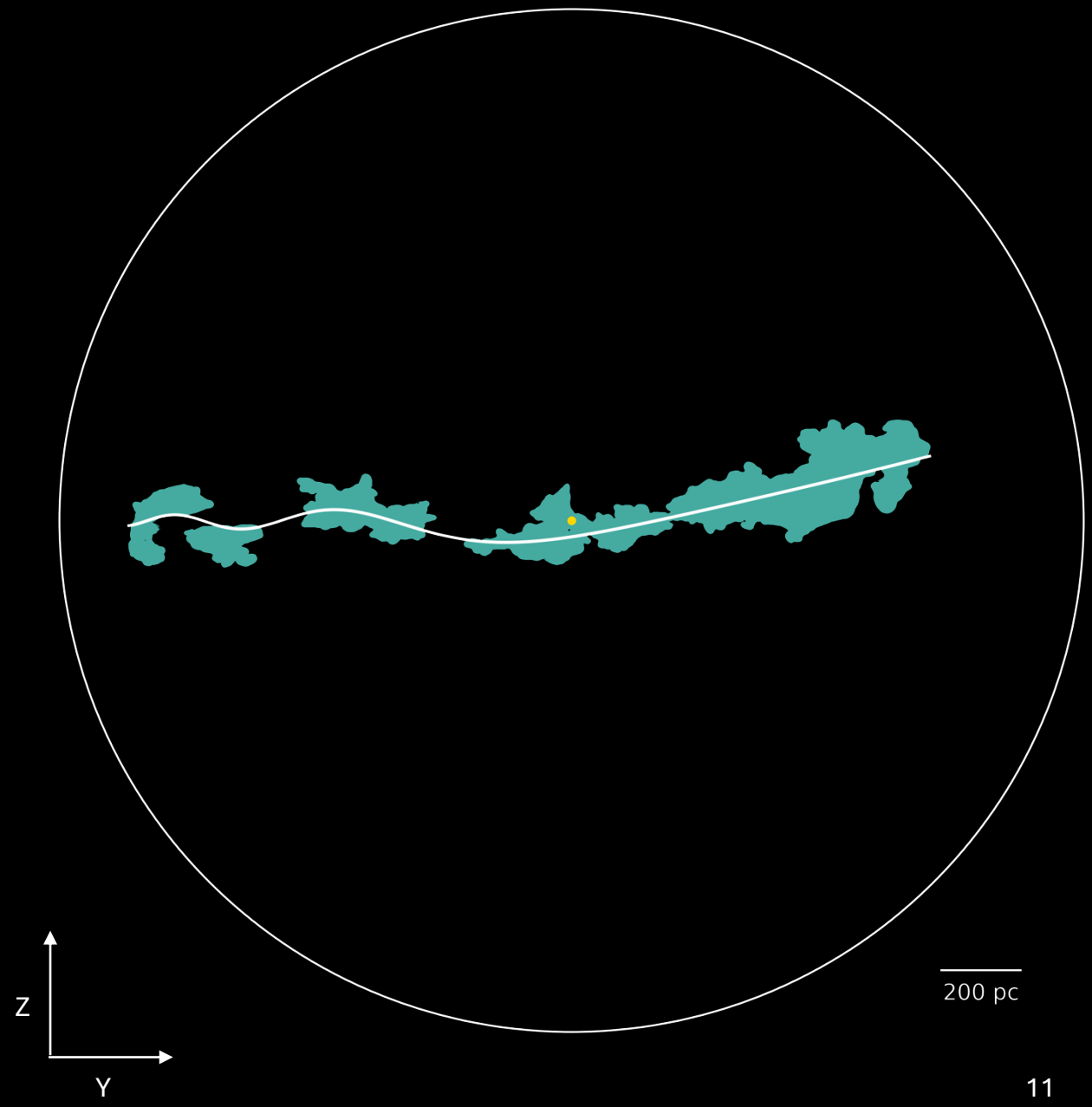
- lengths of 1 – 2.5 kpc
- masses of ~ 10^6 $M_\odot$
- ~ 30° pitch angle



Undulation

- superclouds are rotated by their pitch angle

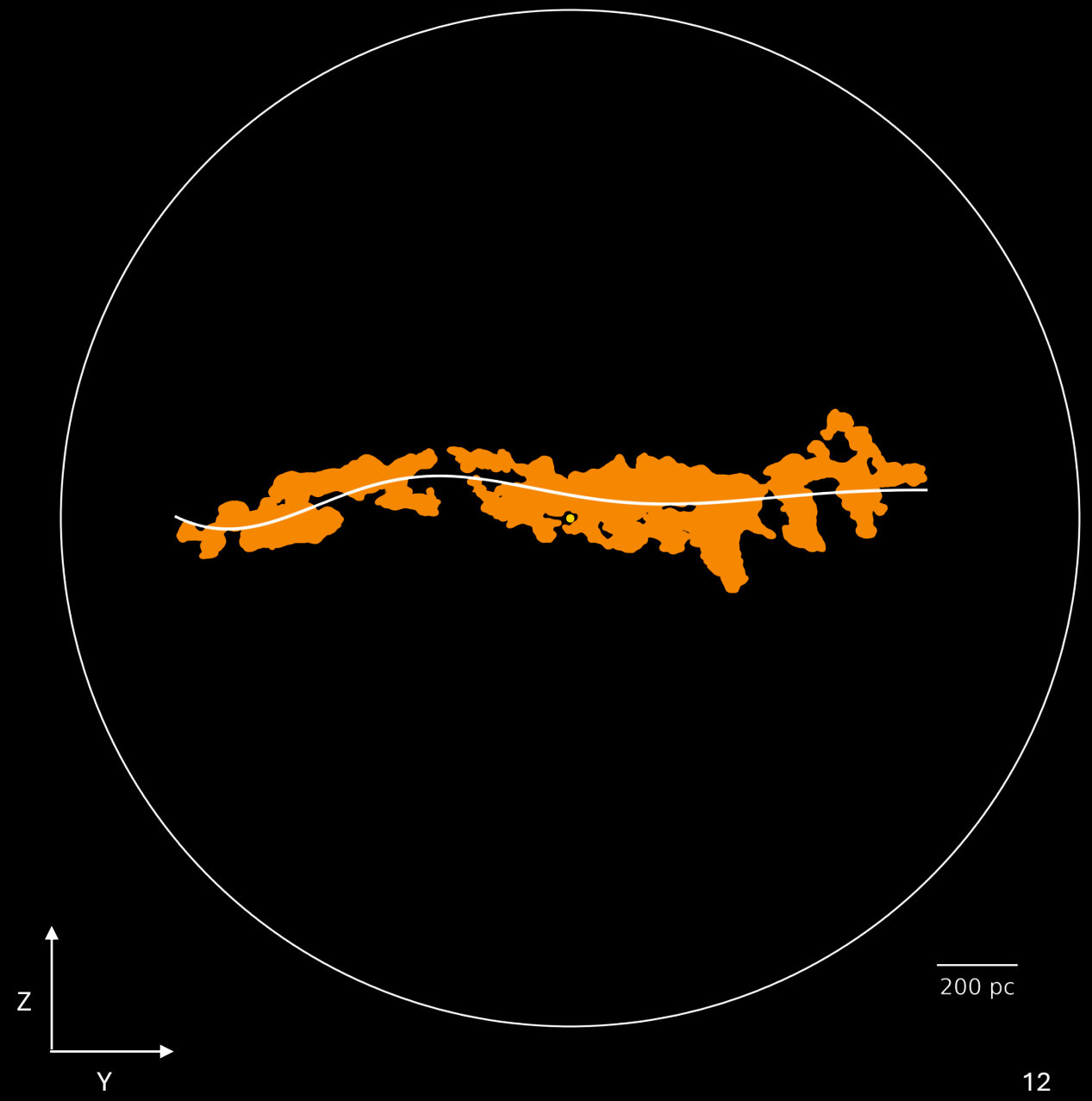
- Natrix Cloud



Undulation

- superclouds are rotated by their pitch angle

- Malpolon Cloud

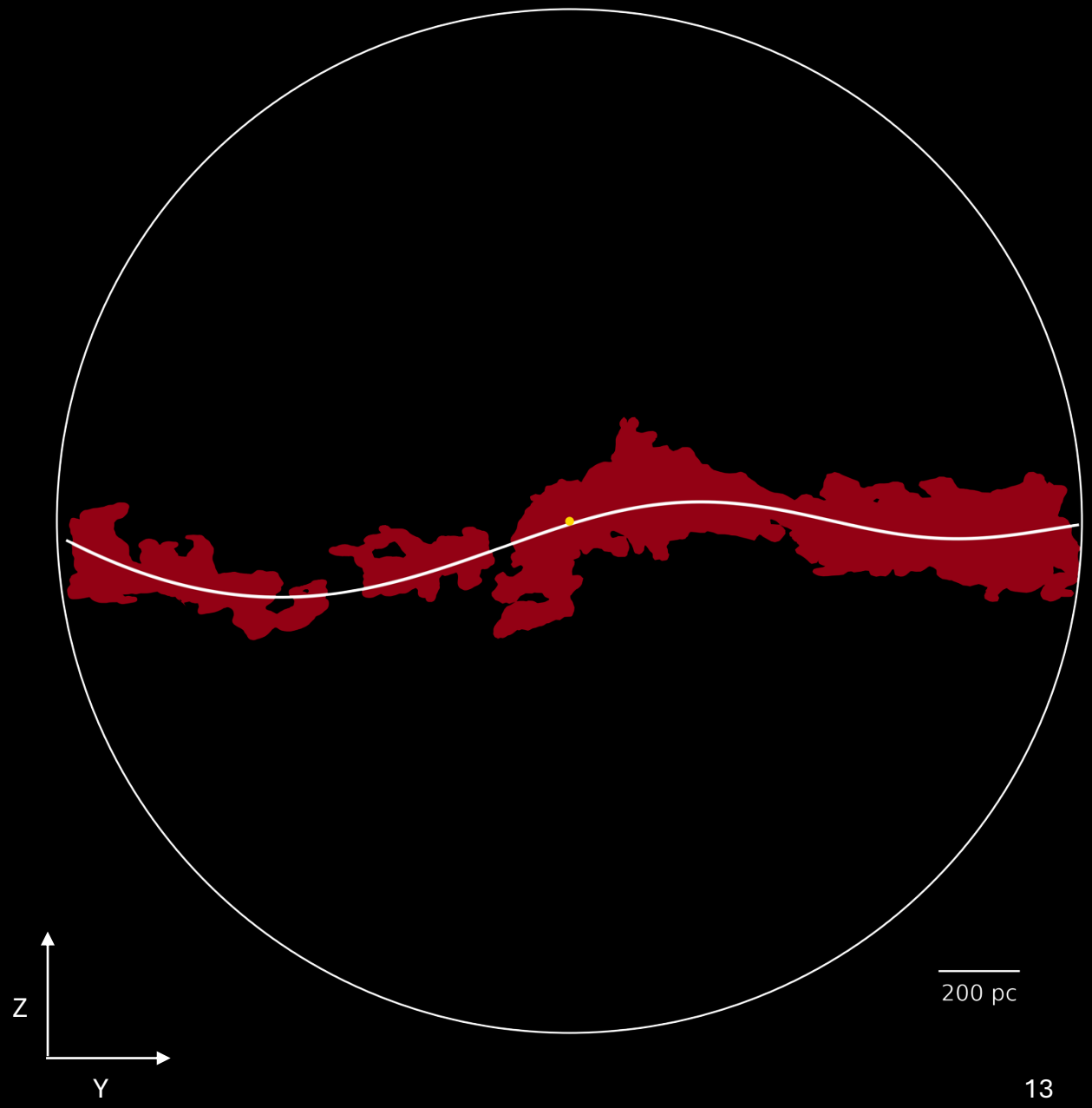


Undulation

superclouds are rotated by their pitch angle

Radcliffe Wave

oscillating structure (Konietzka et al. 2024)

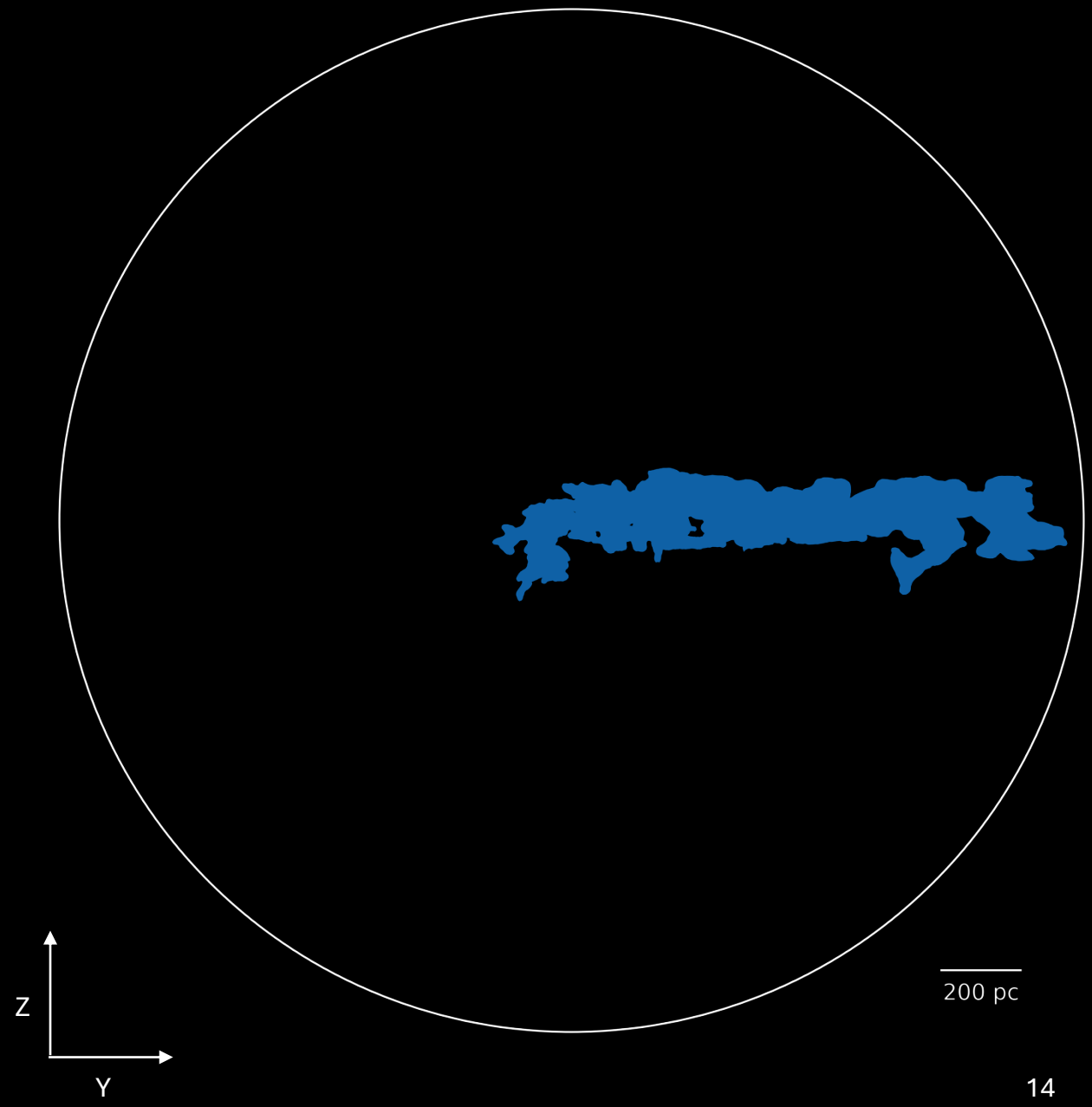


Undulation

superclouds are rotated by their pitch angle

Split

only supercloud showing no undulation



Superclouds

90% of SFRs in the dust map volume inside the superclouds

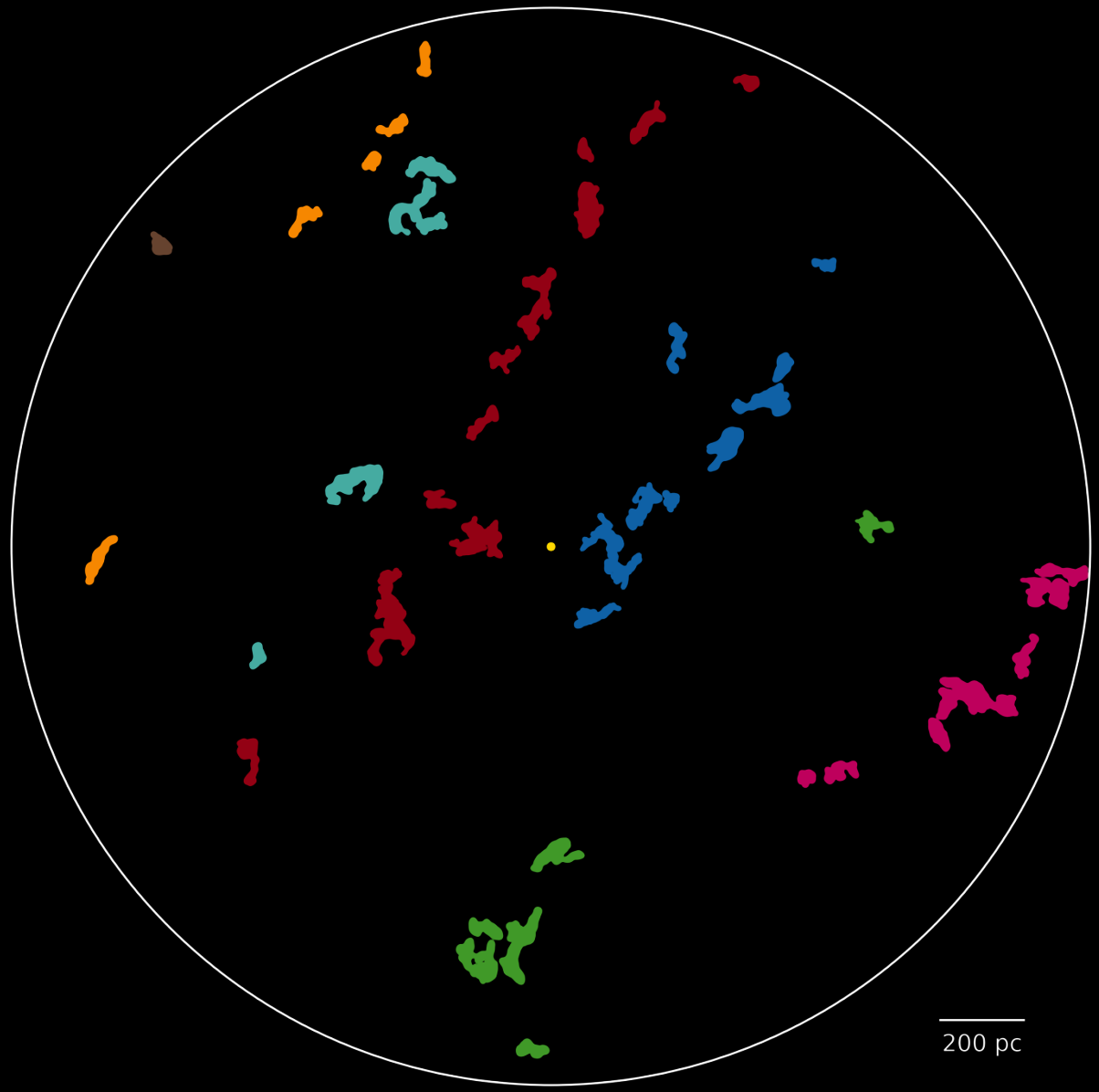
reapplication of HOP using higher density parameters



Superclumps

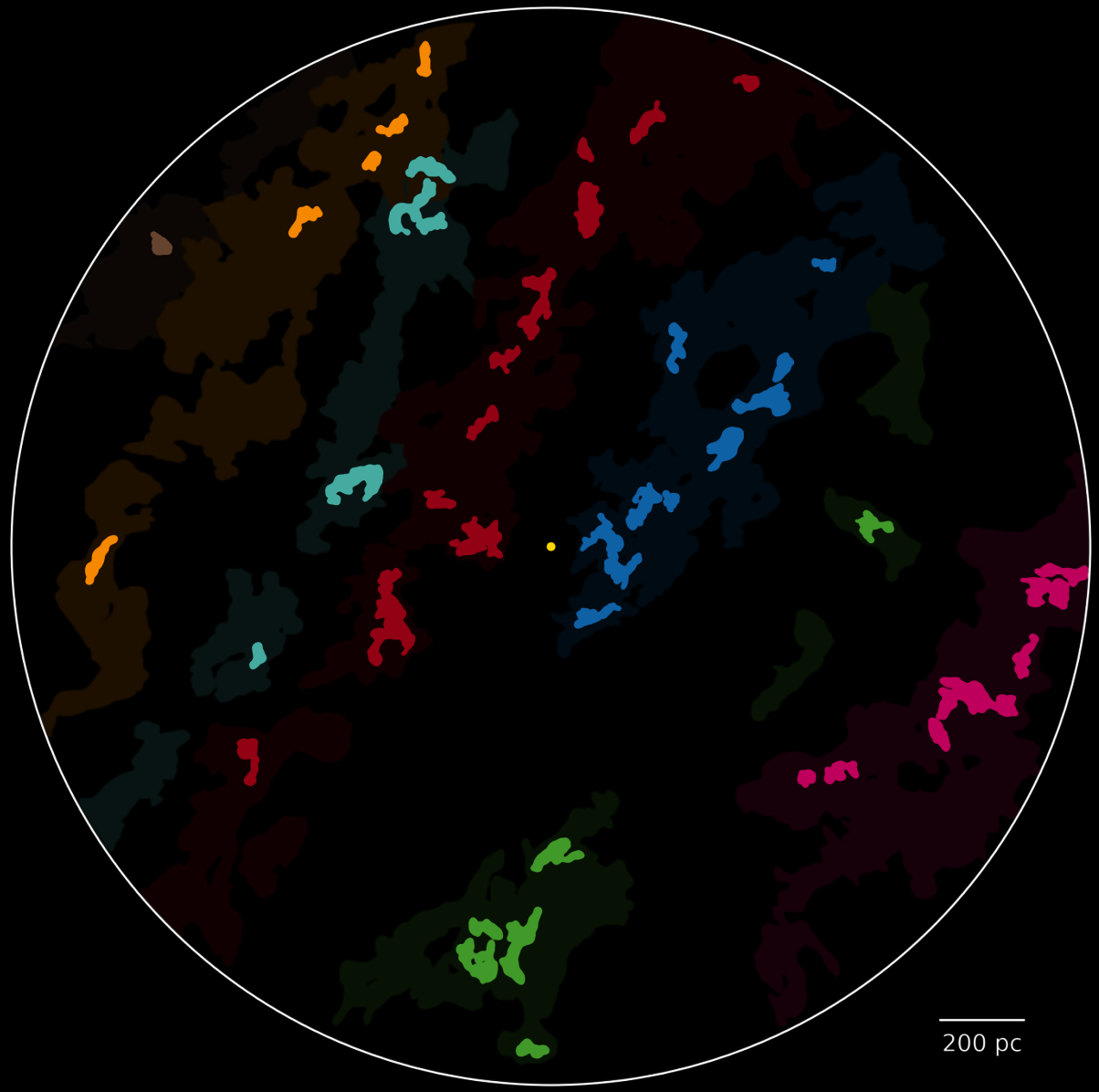
- density enhancements inside the superclouds

- lengths of ~ 100 pc
- masses of ~ 10^4 $M_\odot$
- 150 – 380 pc spacing



Combined View

> superclumps align with supercloud spines

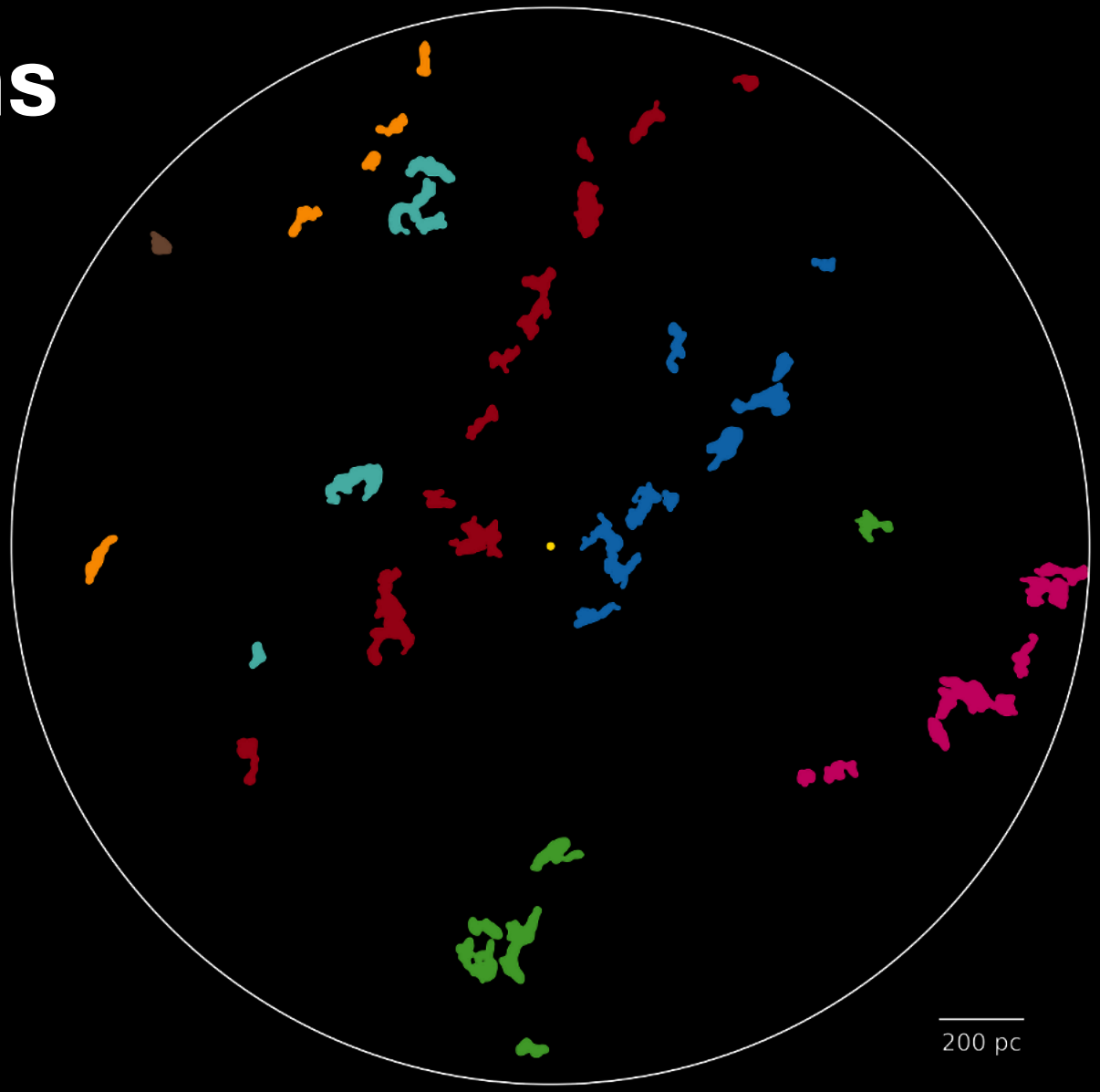


Star-Forming Regions

SFRs from Zucker et al. (2020) within the volume of the dust map

90% within superclouds

73% within superclumps

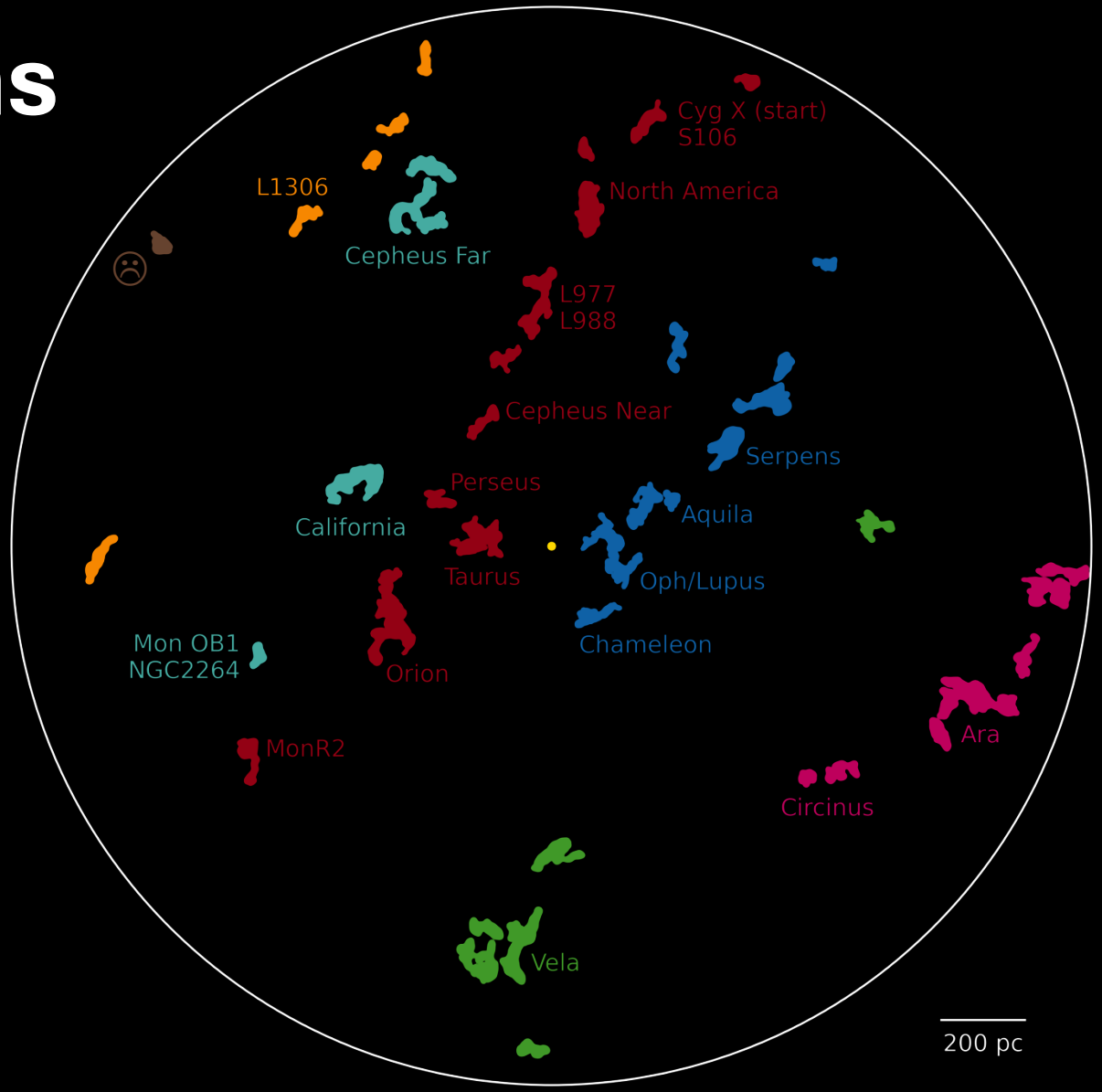


Star-Forming Regions

SFRs from Zucker et al. (2020) within the volume of the dust map

90% within superclouds

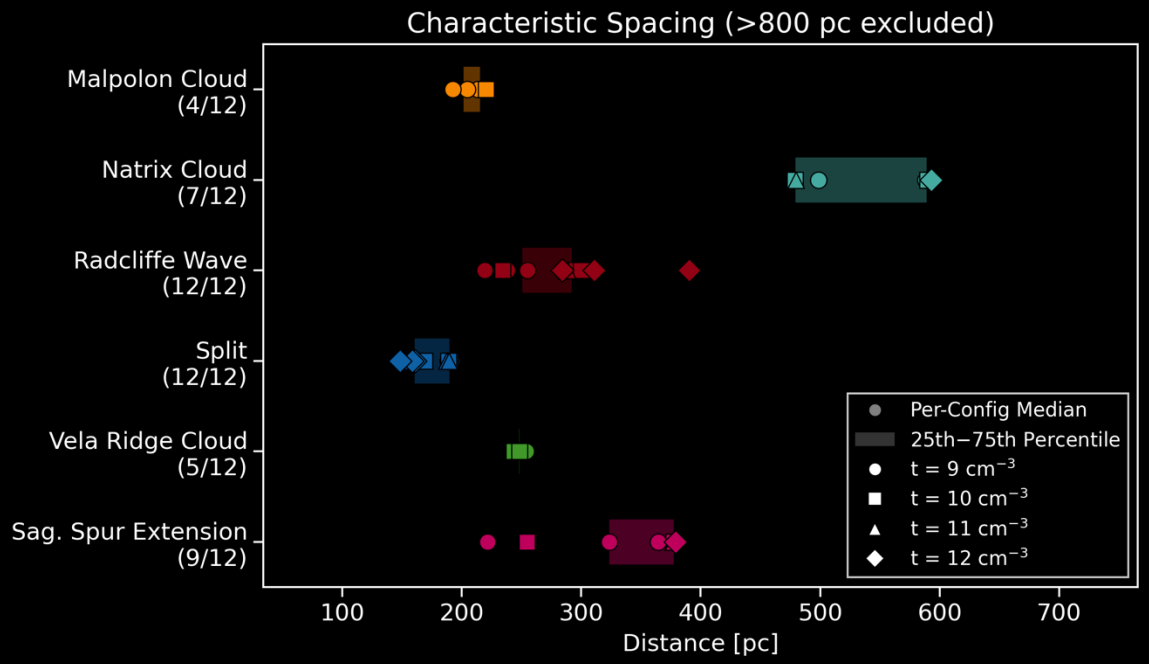
73% within superclumps

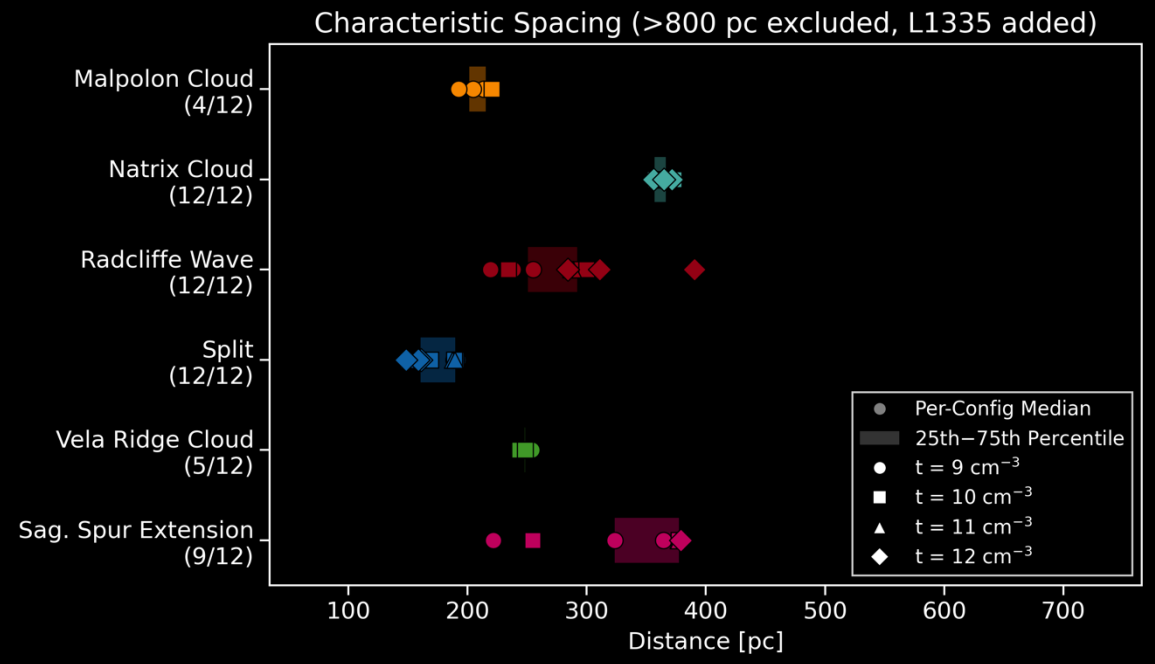


Superclump Spacing

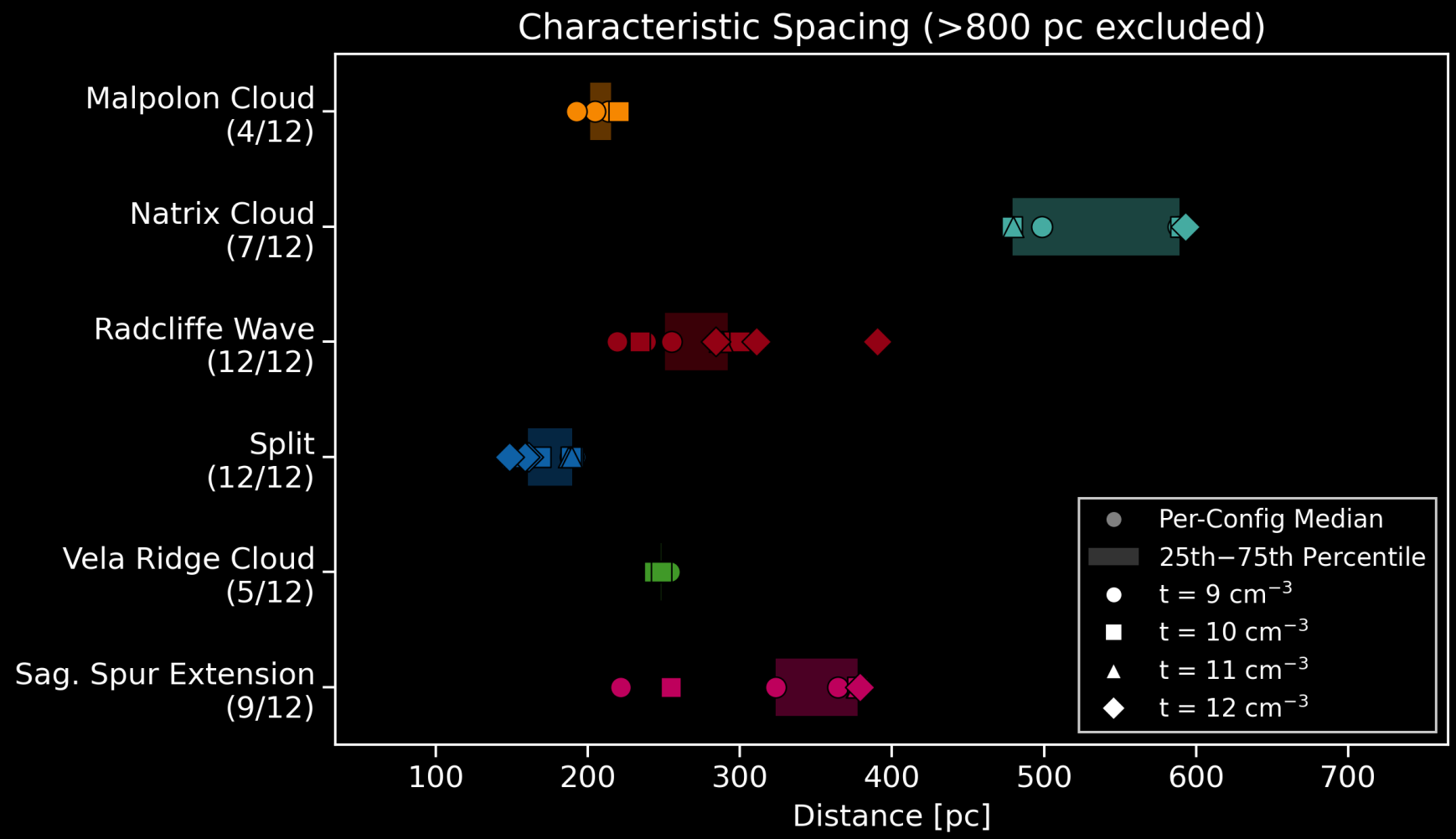
Median spacings for each of the 12 HOP configurations.

! measure of consistency, not a confidence interval

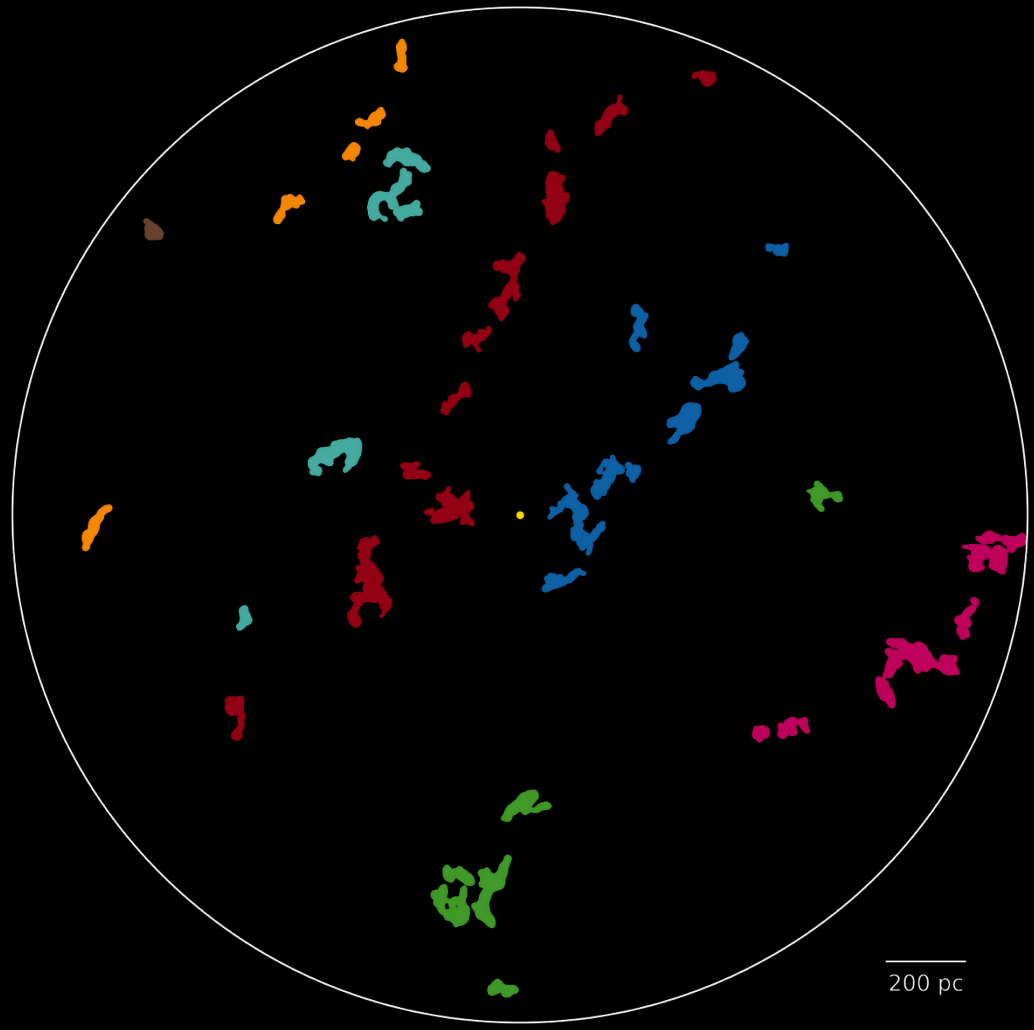




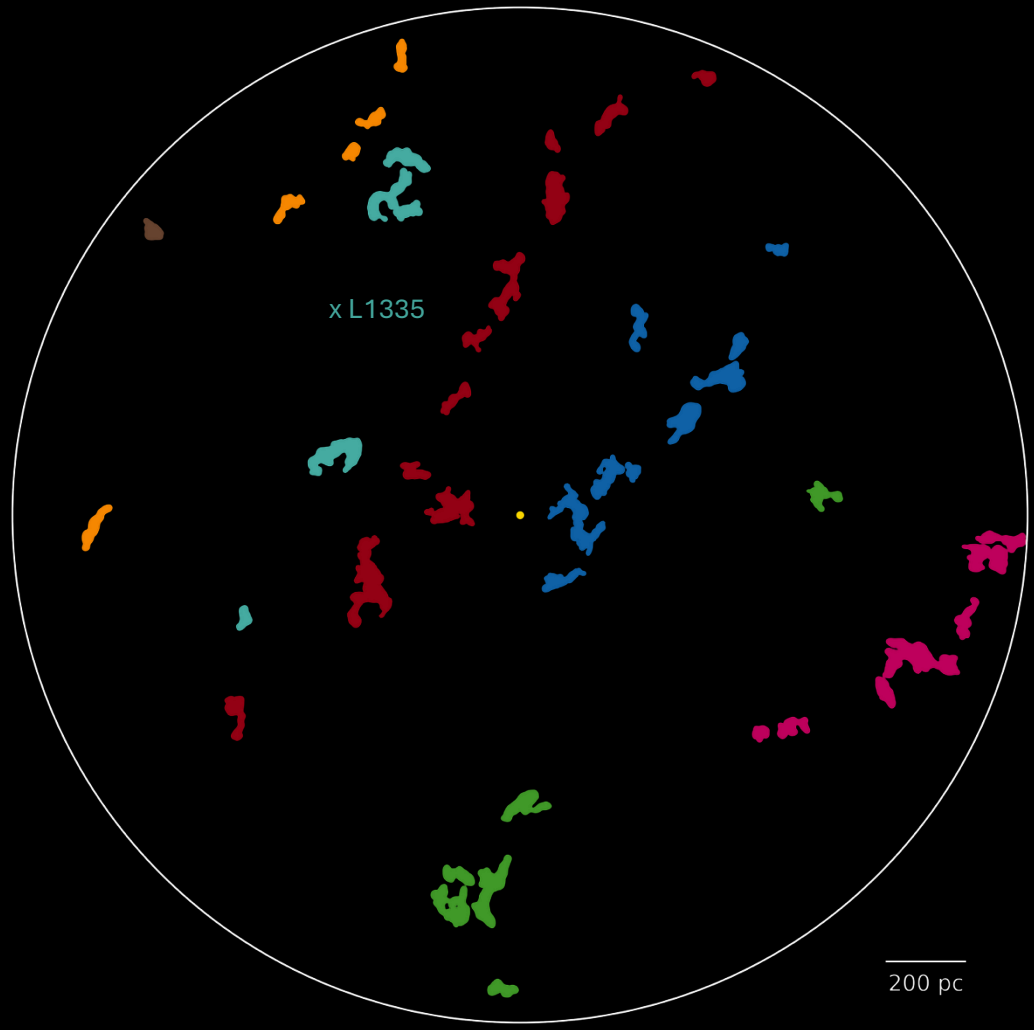
Superclump Spacing



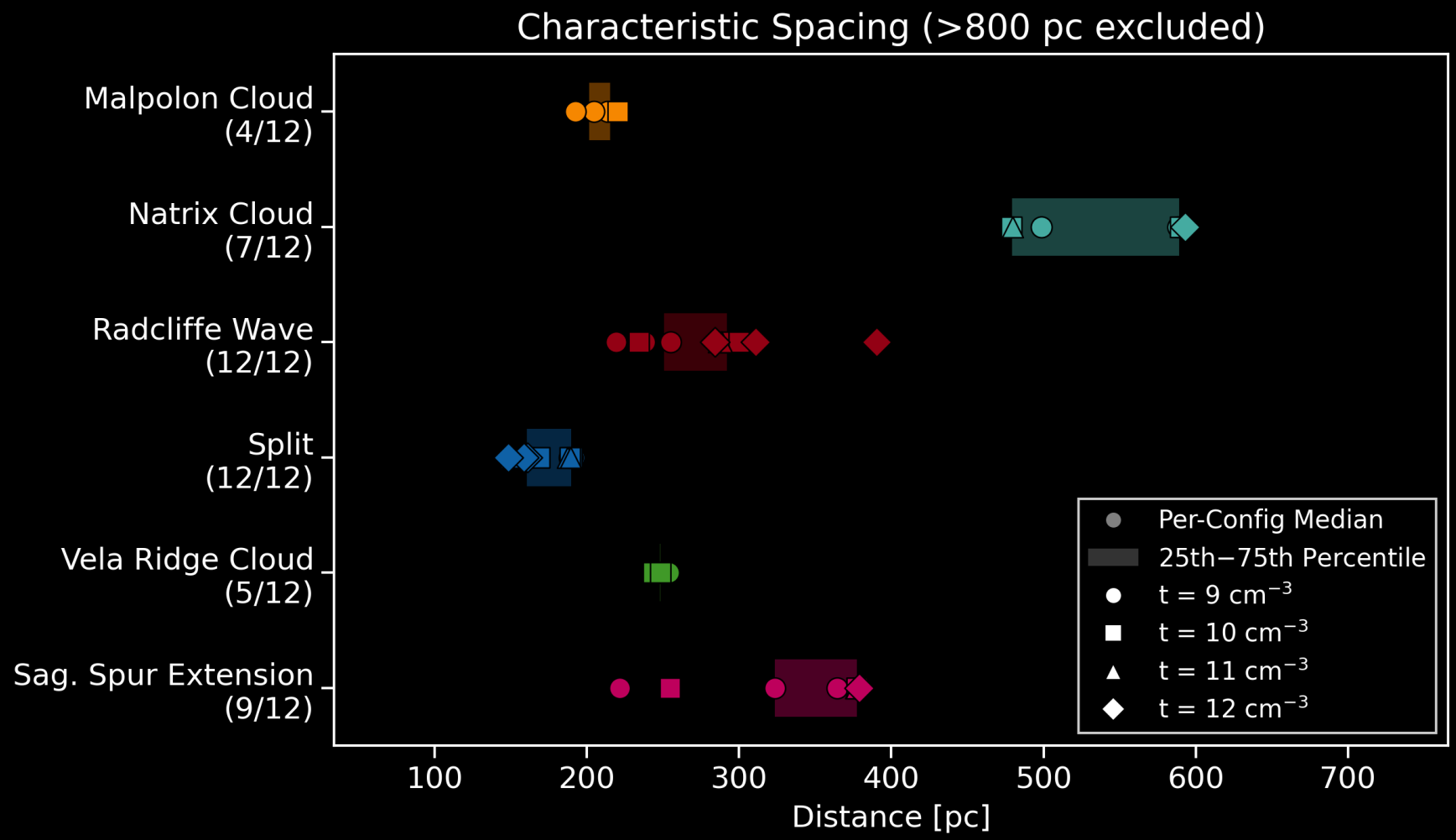
Superclump Spacing



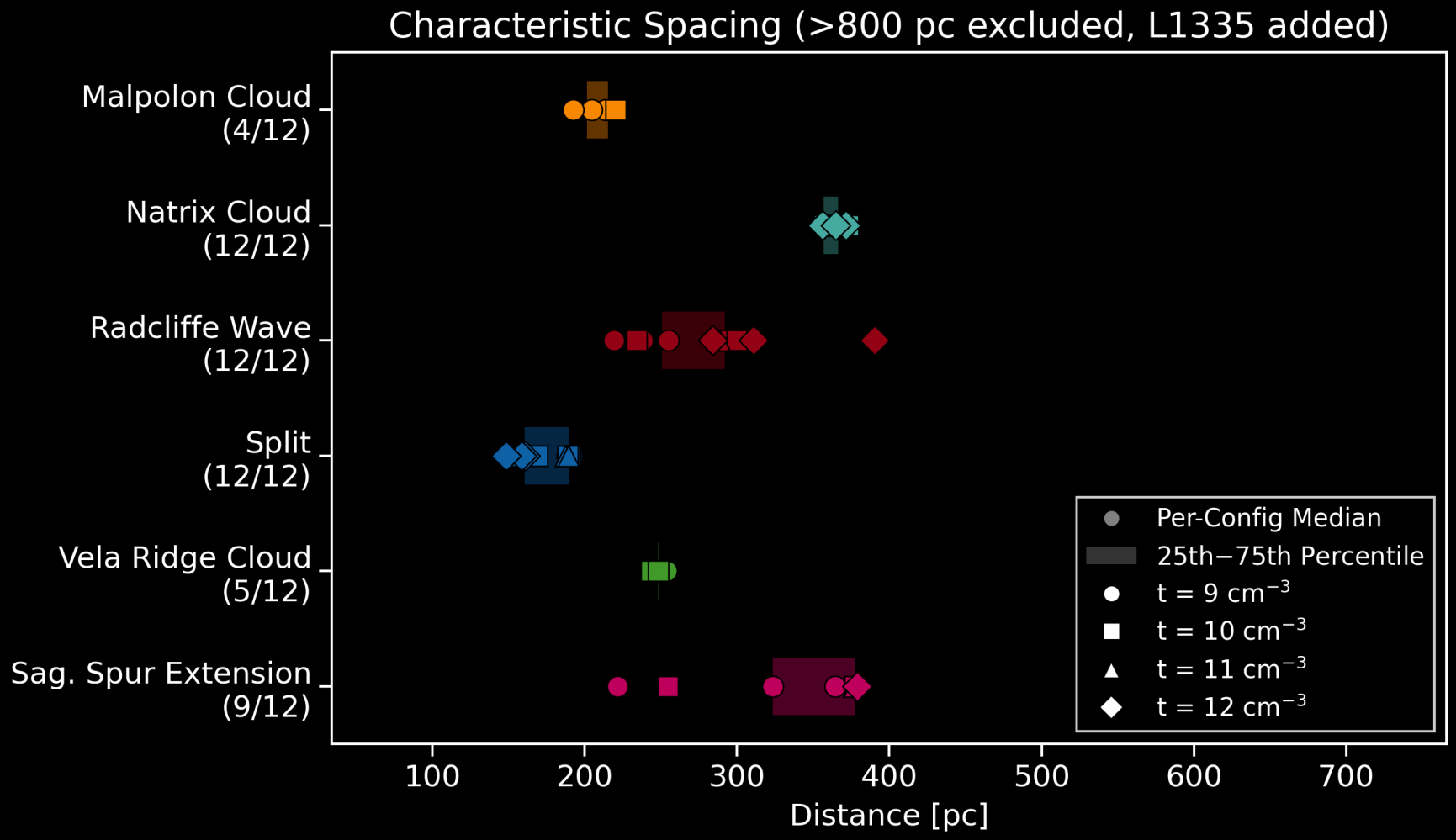
Superclump Spacing



Superclump Spacing



Superclump Spacing



Conclusion

- identification of 7 superclouds in the local Milky Way with 5 previously unknown structures
- inner density analysis reveals superclumps, ~ 73% of known SFRs overlap and we find repeated spacings
- superclouds and superclumps act as bridge between galactic scales down to star-forming regions

Superclouds Kormann+2026

Superclumps (Preprint)